Practice Graded Assignment: Course project

Review fellow learners
Congrats on submitting your assignment! You have reviewed enough peers, but you can help even more learners complete this course by continuing to give reviews.

Review assignments

AI Grading

Your assignment has been graded by AI. View your scores below. You'll also find feedback from Coursera AI, an AI-powered learning assistant, which provides insights on how to enhance your work.

In this project, you will demonstrate your ability to perform the following:

- Create users and groups and implement mandatory multifactor authentication.
- Create a resource group and a new Storage account and provide access to users.
- Enable disk encryption on a Storage account.
- Create a virtual machine and add a network security group rule to allow inbound traffic on port 3389 for your virtual machine.
- Implement Azure security using Defender for Storage and configure diagnostics settings to send data to the Log Analytics workspace.

You will create a deliverable that documents the screenshots of activities related to applying multiple layers of security to protect a confidential design image. The deliverable is the Azure security project solution document. You can create the deliverable using Microsoft Word or a similar application.

We provided a template to help you document your work and create your deliverable in the Project setup reading. If you are yet to download the template, here are the instructions to download the template:

1. Select the template titled Azure security project solution template.

Azure security project solution template
DOCX File

2. Rename the downloaded template with a unique file name so you can easily find it and upload it for submission later. For example, "Azure security project solution_*Your initials*."

3. If you prefer, you can upload the template to a cloud storage service such as OneDrive or Google Drive. You can then open the template using the relevant application. For example, you can open the template in your Google Drive by selecting **Open with** > **Google Docs**.

Your detailed tasks will include the following:

Task 1: Create a group with three users, assign the Virtual Machine Administrator Login role, and implement multifactor authentication for the users.

- Create three users in Azure Active Directory (Azure AD).
- Create a group for the three users and assign the Virtual Machine Administrator Login role to this group.
- Implement mandatory multifactor authentication for all three users.

Task 2: Upload an image to a Storage account container, create a Blob shared access signature (SAS) URL for the image, and assign the Storage Blob Data Contributor role to the group members.

- Create a resource group, a Storage account, and a container.
- Upload an image to the container in the Storage account and create a Blob SAS URL for the image.
- Assign the Storage Blob Data Contributor role to the three members of the group.

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Task 3: Create a Key Vault and a managed identity and encrypt the Storage account with customer-managed keys.

- Create a Key Vault and a managed identity and assign key permissions to the managed identity.
- Encrypt your Storage account using customer-managed keys.

Task 4: Secure your Storage account using Microsoft Defender and Log Analytics workspace.

- Enable Microsoft Defender for Storage for your Storage account.
- Configure diagnostic settings to send data to the Log Analytics workspace.

Task 5: View the image using a Blob SAS URL on a virtual machine.

- Create a Windows Server 2019 Datacenter virtual machine.
- Add a network security group rule to allow inbound traffic on port 3389 for your virtual machine.
- Verify access to the image as an administrator using the Blob SAS URL.
- Access the image as a sales team member using the Blob SAS URL.

At the end of this project, you will have your version of the following deliverable for submission:

- *Azure security project solution.docx*

As part of this project, you will also review the project submissions of two peers using the peer review rubric provided.

You should plan to complete this course project in about 1.5 hours.

Step-By-Step Assignment Instructions

Project scenario

The activities in this project will focus on safeguarding the intellectual property of BuyforSure. You will protect a confidential design image that the company will use for marketing. You will apply multiple layers of security to ensure that only three new sales team members can access the image and that it is protected from malicious attacks.

Completing the project tasks: Task 1 of 5

Creating a well-formatted document with screenshots will help your peer effectively review your submission. Follow the instructions below to create your project deliverable.

Task 1: Create a group with three users, assign the Virtual Machine Administrator Login role, and implement multifactor authentication for the users.

Activity 1: Create three users in Azure Active Directory (Azure AD).

While performing this activity, take a screenshot of the result(s) and add it to your *Azure security project solution* document.

In this activity, you will create three new users, AlexSmith, SofiaLee, and NishaPatel, in Azure AD. You can use the following information while performing this activity:

Note: It is not mandatory that you follow the specific names or settings. However, using these details will make it easier to conduct peer reviews or compare your work with the exemplar that you'll download at the end of this project. If you need to retake a screenshot and the previous value is not accepted, you can suffix a number to the value. For example, in the place of AlexSmith, you can use AlexSmith1.

Field	Value
User principal name	**AlexSmith1** Select any domain.
Mail nickname	Select the option to derive from the user principal name.
Display name	**AlexSmith**
Password	Auto-generate the password.
Account enabled	Select the **Account enabled** checkbox.

Repeat the steps for creating the other two users: **SofiaLee** and **NishaPatel**.

Remember to include the following screenshot to help your peers evaluate:

- A screenshot that shows the three users on the **Users** page

Activity 2: Create a group for the three users and assign the Virtual Machine Administrator Login role to this group.

While performing this activity, take a screenshot of the result(s) and add it to your *Azure security project solution* document.

In this activity, you will create a group called **JewelSales** to add the three sales team members. You will also provide a meaningful description for this group. You will also assign the **Virtual Machine Administrator Login** role for the group. You can use the following information while performing this activity:

Note: It is not mandatory that you follow the specific names or settings. However, using these details will make it easier to conduct peer reviews or compare your work with the exemplar that you'll download at the end of this project.

Field	Value
Group type	Select **Microsoft 365**.
Group name	**JewelSales**
Group email address	Keep the automatically populated email address.
Group description	**Sales team responsible for promoting and selling the jewelry brand**

Remember to include the following:

- A screenshot that shows the three users on the **All groups** page
- A screenshot that shows the **Added Role assignment** notification indicating that the group was assigned the Azure **Virtual Machine Administrator Login** role

Activity 3: Implement mandatory multifactor authentication for all three users.

While performing this activity, take a screenshot of the result(s) and add it to your *Azure security project solution* document.

Remember to include the following:

- A screenshot that shows the **MULTI-FACTOR AUTH STATUS** enabled for all three users

Completing the project tasks: Task 2 of 5

Task 2: Upload an image to a Storage account container, create a Blob shared access

Task 2: Upload an image to a Storage account container, create a Blob shared access signature (SAS) URL for the image, and assign the Storage Blob Data Contributor role to the group members.

Activity 1: Create a resource group, a Storage account, and a container.

While performing this activity, take a screenshot of the result(s) and add it to your *Azure security project solution* document.

In this activity, you will create a resource group called **JewelPromo**. Then, you will create a Storage account. Choosing a meaningful name like **advtstorage** for the Storage account will indicate that it is used for advertising and marketing content. You can use the following information while performing this activity:

Note: It is not mandatory that you follow the specific names or settings. However, using these details will make it easier to conduct peer reviews or compare your work with the exemplar that you'll download at the end of this project.

Resource group

Field	**Value**
Subscription	Select your Azure subscription.
Resource group	**JewelPromo**
Region	**East US**

Storage account

Field	**Value**
Subscription	Select the Azure subscription for the new Storage account.
Resource group	**JewelPromo**
Storage account name	**advtstorage**
Region	East US
Performance	**Standard: Recommended for most scenarios (general-purpose v2 account)**
Redundancy	**Geo-redundant storage (GRS).** Note that read access to data is available in the event of regional unavailability.

Container

Field	**Value**
Container name	**campaignimages**
Public access level	**Blob (anonymous read access for blobs only)**

Remember to include the following:

- A screenshot that shows the new resource group listed on the **Resource groups** page
- A screenshot that shows the **Containers** page of the new resource group with the new container listed

Activity 2: Upload an image to the container in the Storage account and create a Blob SAS URL for the image.

While performing this activity, take a screenshot of the result(s) and add it to your *Azure security project solution* document.

In this activity, you will upload the confidential design image called **SparklingGems.png** to the container. You can have this image saved in your local system. After you create the Blob SAS URL for the image, **remember to save the URL for later use**.

You can use the following information while performing this activity:

Note: It is not mandatory that you follow the specific names or settings. However, using these details will make it easier to conduct peer reviews or compare your work with the exemplar that you'll download at the end of this project.

Uploading the image

Field	Value
Image name	**SparklingGems.png**

Creating Blob SAS URL

Field	Value
Permissions	**Read**, **Write**
Start	You may want to choose an immediate start time and date
Expiry	You may want to choose an expiry time that allows you to successfully complete this project.

Remember to include the following:

- A screenshot that shows the uploaded image on the Storage account's **Containers** page
- A screenshot that shows the Blob SAS URL for the image

Activity 3: Assign the Storage Blob Data Contributor role to the three members of the group.

While performing this activity, take a screenshot of the result(s) and add it to your *Azure security project solution* document.

Remember to include the following:

- A screenshot that shows the **Storage Blob Data Contributor** role assigned to the three users on the **Access control (IAM)** page of **Resource groups**

Completing the project tasks: Task 3 of 5

Task 3: Create a Key Vault and a managed identity and encrypt the Storage account with customer-managed keys.

Activity 1: Create a Key Vault and a managed identity and assign key permissions to the managed identity.

While performing this activity, take a screenshot of the result(s) and add it to your *Azure security project solution* document.

In this activity, you will create a Key Vault called **AdKeyVault1** and a managed identity called **CampaignIdentity** with the required permissions. You can use the following information while performing this activity:

Note: It is not mandatory that you follow the specific names or settings. However, using these details will make it easier to conduct peer reviews or compare your work with the exemplar that you'll download at the end of this project.

Key Vault

Field	Value
Subscription	Select an Azure subscription.
Resource group	**JewelPromo**
Key vault name	**AdKeyVault1**
Region	**East US**
Pricing tier	**Standard**
Purge protection	Enable purge protection

Managed identity

Field	Value
Subscription	Select an Azure subscription.
Resource group	**JewelPromo**
Region	**East US**
Name	**CampaignIdentity**

Permissions

Field	Value
Key Management Operations	**Get**, **List**
Cryptographic Operations	**Unwrap Key**, **Wrap Key**

Remember to include the following:

- A screenshot that shows a notification that the Key Vault is successfully deployed
- A screenshot that shows a notification that the managed identity is successfully deployed
- A screenshot that shows that key permissions are assigned to the managed identity

Activity 2: Encrypt your Storage account using customer-managed keys.

While performing this activity, take a screenshot of the result(s) and add it to your *Azure security project solution* document.

In this activity, you will create a customer-managed key called **AdKey**. You can use the following information while performing this activity:

Note: It is not mandatory that you follow the specific names or settings. However, using these details will make it easier to conduct peer reviews or compare your work with the exemplar that you'll download at the end of this project.

Field	Value
Storage account	**advtstorage**

Key Vault	**AdKeyVault1**
Key	**AdKey**
Managed identity	**Campaignidentity**

Remember to include the following:

- A screenshot that shows a notification that the Storage account is successfully encrypted using customer-managed keys

Completing the project tasks: Task 4 of 5

Task 4: Secure your Storage account using Microsoft Defender and Log Analytics workspace.

Activity 1: Enable Microsoft Defender for Storage for your Storage account.

While performing this activity, take a screenshot of the result(s) and add it to your *Azure security project solution* document.

In this activity, you will secure your Storage account, **advtstorage**.

Note: It is not mandatory that you follow the specific names or settings. However, using these details will make it easier to conduct peer reviews or compare your work with the exemplar that you'll download at the end of this project.

Remember to include the following:

- A screenshot that shows the **Microsoft Defender for Storage** enablement status as **On**

Activity 2: Configure diagnostic settings to send data to the Log Analytics workspace.

While performing this activity, take a screenshot of the result(s) and add it to your *Azure security project solution* document.

In this activity, you will configure a diagnostic setting called **AdStorageMonitor** for your Storage account **advtstorage**. You will ensure that transaction metrics are sent to the Log Analytics workspace. You can use the following information while performing this activity:

Note: It is not mandatory that you follow the specific names or settings. However, using these details will make it easier to conduct peer reviews or compare your work with the exemplar that you'll download at the end of this project.

Diagnostic setting

Field	Value
Diagnostic setting name	**AdStorageMonitor**

Remember to include the following:

- A screenshot that shows the new diagnostic setting on the **Diagnostic settings** page

Completing the project tasks: Task 5 of 5

Task 5: View the image using a Blob SAS URL on a virtual machine.

Activity 1: Create a Windows Server 2019 Datacenter virtual machine.

While performing this activity, take a screenshot of the result(s) and add it to your *Azure security project solution* document.

In this activity, you will create a virtual machine called **CampaignVM**. You can use the following information while performing this activity:

Note: It is not mandatory that you follow the specific names or settings. However, using these details will make it easier to conduct peer reviews or compare your work with the exemplar that you'll download at the end of this project.

Field	Value
Subscription	Select an Azure subscription.
Resource group	**JewelPromo**
Virtual machine name	**CampaignVM**
Region	East US
Availability options	**Availability zone**
Availability zone	**Zones 1**
Administrator account username	**VMAdmin**
Public inbound ports	**None**
Virtual network	**CampaignVM-vnet**
Subnet	Select the default value.
NIC network security group	**Basic**
Boot diagnostics	**Disable**

Remember to include the following:

- A screenshot that shows the **Your deployment is complete** message indicating that your virtual machine has been created

Activity 2: Add a network security group rule to allow inbound traffic on port 3389 for your virtual machine.

While performing this activity, take a screenshot of the result(s) and add it to your *Azure security project solution* document.

In this activity, you will add a network security group named **AllowAnyCustom3389Inbound**. You can use the following information while performing this activity:

Note: It is not mandatory that you follow the specific names or settings. However, using these details will make it easier to conduct peer reviews or compare your work with the exemplar that you'll download at the end of this project.

Field	Value
Service	**Custom**
Destination port ranges	**3389**
Protocol	**Any**
Action	**Allow**
Priority	**310**
Name	**AllowAnyCustom3389Inbound**

Remember to include the following:

- A screenshot that shows the new inbound port rule added for port 3389.

Activity 3: Verify access to the image as an administrator using the Blob SAS URL.

While performing this activity, take a screenshot of the result(s) and add it to your *Azure security project solution* document.

In this activity, you will use your administrator login **VMAdmin** to verify the Blob SAS URL. You can use the following information while performing this activity:

Note: It is not mandatory that you follow the specific names or settings. However, using these details will make it easier to conduct peer reviews or compare your work with the exemplar that you'll download at the end of this project.

Field	**Value**
User name	**VMAdmin**
Blob SAS URL	Use the URL you saved in Task 2 Activity 2

Remember to include the following:

- A screenshot that shows the image in a virtual machine browser with the Blob SAS URL

Activity 4: Access the image as a sales team member using the Blob SAS URL.

While performing this activity, take a screenshot of the result(s) and add it to your *Azure security project solution* document.

In this activity, you will sign in to the Azure portal as a user, for example, **SofiaLee,** to verify the Blob SAS URL. You can use the following information while performing this activity:

Note: It is not mandatory that you follow the specific names or settings. However, using these details will make it easier to conduct peer reviews or compare your work with the exemplar that you'll download at the end of this project.

Field	**Value**
Azure portal login	**SofiaLee**
Virtual machine login	**VMAdmin**
Blob SAS URL	Use the URL you saved in Task 2 Activity 2

Remember to include the following:

- A screenshot that shows the Connect page of the virtual machine with one of the sales team members logged in
- A screenshot that shows the image in a virtual machine browser with the Blob SAS URL

Finalizing your assignment

After completing all the tasks, you can finalize your deliverables. Ensure that you've added the steps and screenshots for all the tasks.

Remove any instruction sections or examples provided in the original deliverable templates. Your final deliverables should contain only your work. You are then ready to upload your deliverables in the **My submission** tab.

Review criteria

Twenty points are allotted for this capstone project. The points are distributed across the various tasks and activities that you have completed while developing your version of the Azure security project solution document.

Your grade will be based on the following distribution of points across tasks and activities.

Deliverable: Azure security project solution

Task 1: Create a group with three users, assign the Virtual Machine Administrator Login role, and implement multifactor authentication for the users. (4 pts.)

Activity 1: Create three users in Azure Active Directory (Azure AD). (1 pt.)

Activity 2: Create a group for the three users and assign the Virtual Machine Administrator Login role to this group. (2 pts.)

Activity 3: Implement mandatory multifactor authentication for all three users. (1 pt.)

Task 2: Upload an image to a Storage account container, create a Blob shared access signature (SAS) URL for the image, and assign the Storage Blob Data Contributor role to the group members. (5 pts.)

Activity 1: Create a resource group, a Storage account, and a container. (2 pts.)

Activity 2: Upload an image to the container in the Storage account and create a Blob SAS URL for the image. (2 pts.)

Activity 3: Assign the Storage Blob Data Contributor role to the three members of the group. (1 pt.)

Task 3: Create a Key Vault and a managed identity and encrypt the Storage account with customer-managed keys. (4 pts.)

Activity 1: Create a Key Vault and a managed identity and assign key permissions to the managed identity. (3 pts.)

Activity 2: Encrypt your Storage account using customer-managed keys. (1 pt.)

Task 4: Secure your Storage account using Microsoft Defender and Log Analytics workspace. (2 pts.)

Activity 1: Enable Microsoft Defender for Storage for your Storage account. (1 pt.)

Activity 2: Configure diagnostic settings to send data to the Log Analytics workspace. (1 pt.)

Task 5: View the image using a Blob SAS URL on a virtual machine. (5 pts.)

Activity 1: Create a Windows Server 2019 Datacenter virtual machine. (1 pt.)

Activity 2: Add a network security group rule to allow inbound traffic on port 3389 for your virtual machine. (1 pt.)

Activity 3: Verify access to the image as an administrator using the Blob SAS URL. (1 pt.)

Activity 4: Access the image as a sales team member using the Blob SAS URL. (2 pts.)

Preparing for project submission

Note: Do not proceed to the **My submission** and **Peers to review** tabs until you've completed your version of the following deliverable:

- *Azure security project solution*

Review, proofread, and edit your deliverables. Ensure they are ready to submit for peer review.

To submit the deliverables, upload your documents to the Coursera platform following the prompts in the **My submission** and **Peers to review** tabs.

Peer review

Giving, receiving, and incorporating feedback is a crucial professional skill. After you submit your deliverable, as the next activity, you will have to review and provide feedback for two peers' submissions. Similarly, your submission will be reviewed by a peer. We've shared a rubric to ease the review process. You will be presented with questions about your peers' submissions. For example, did your peer document all the key evidence? If prompted, you may have to provide more in-depth, written feedback for some questions. Please provide constructive feedback to help your peers understand what they did well and what they can improve on.

Compare your work (Discussion)

After submitting your deliverables and completing the two peer reviews, you will be able to access the Azure security project solution exemplar. For your ease of reference, the exemplar has been split into two parts:

- *Azure security project solution_Part 1*
- *Azure security project solution_Part 2*

These exemplars are step-by-step guides for the project tasks. You can compare the exemplars with your submissions. You can then choose to reflect on your learning and share your assignment experience with peers in the discussion forum.

After completion of the project

When working on your Azure subscription, at the end of a project, it's good practice to assess whether you still require the resources you created. Running resources can cost you money. You can delete resources individually or delete the entire resource group containing the resources. Here's how you can do this.

Clean up resources

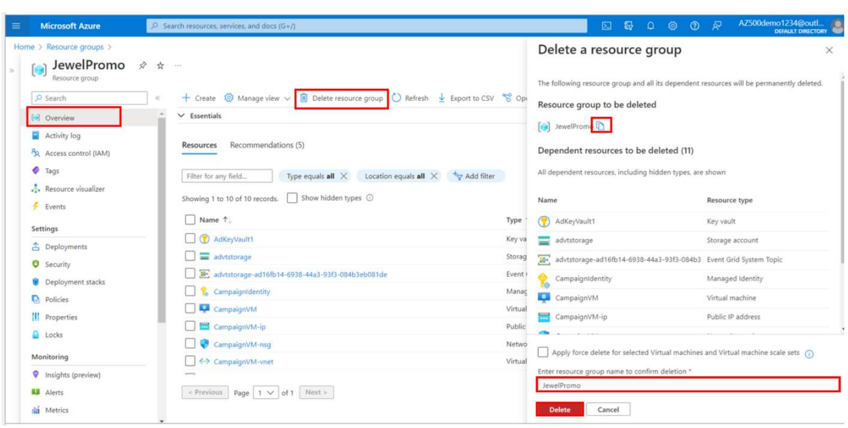
1. In the Azure portal, navigate to **Resource groups** to view the resource group you created for this project. For example, **JewelPromo**.

2. Select the resource group.

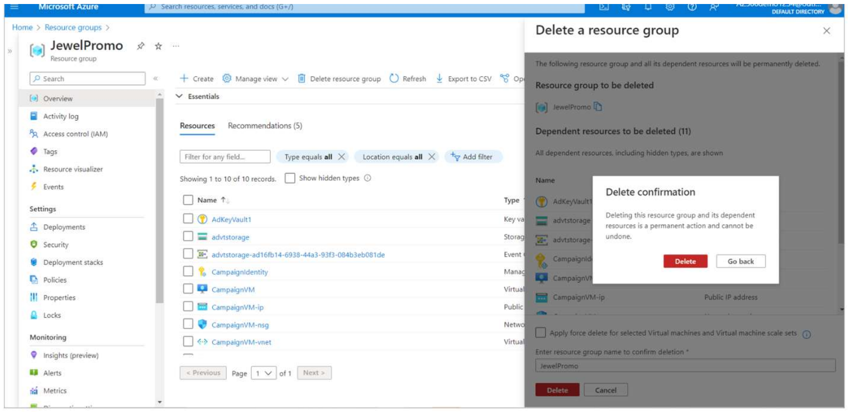
3. On the selected resource group's **Overview** page, select **Delete resource group** to delete the entire resource group.

4. In the **Delete a resource group** wizard that appears, copy and paste the resource group name in the **Enter resource group name to confirm deletion** box.

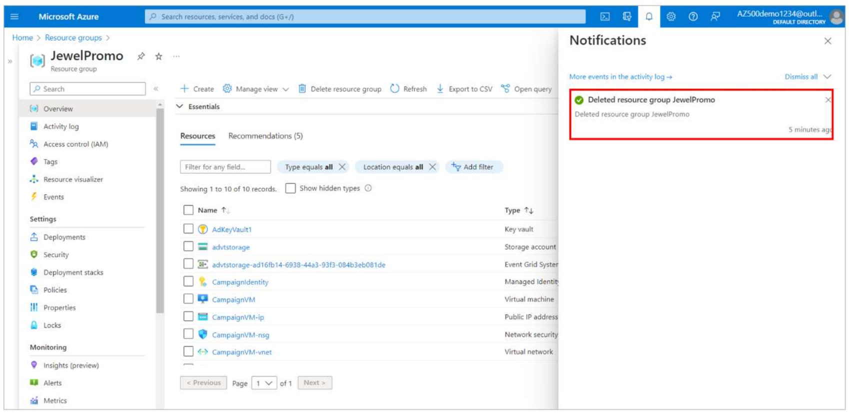
5. Select **Delete**.



6. Then, in the **Delete confirmation** pop-up, select **Delete**.



7. A **Deleted resource group JewelPromo** pop-up window will appear once the resource group is deleted successfully.



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